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Chapter 1

Introduction

Lecture 1: General Overview

08:10 AM Thu, Sep 25 2025

Let $E \neq \emptyset$ a set.

A binary operation $\cdot$ on E is any map from $E \times E$ into E ,

$$\begin{aligned}(\cdot) : E \times E &\longrightarrow E \\(x, y) &\longmapsto x \cdot y\end{aligned}$$

Let $A \subset E$, we say A is a stable by $(\cdot)$ if $(\cdot)$ is also a Binary Operation on A ,

$$\begin{aligned}(\cdot_A) : A \times A &\longrightarrow A \\(x, y) &\longmapsto x \cdot_A y = x \cdot y\end{aligned}$$

Definition 1.0.1 (Group) : Let $G \neq \emptyset$ a set with a Binary Operation $(*)$, we say that G is a group if :

1. $(*)$ is associative, if :

$$\forall x, y, z \in G : (x * y) * z = x * (y * z)$$

2. $(*)$ admits a netural elements if :

$$\exists e \in G, \forall x \in G : x * e = e * x = x$$

- 3.

$$\forall x \in G, \exists x' \in G : x * x' = x' * x = e$$

if $(*)$ is commutative i.e.:

$$\forall x, y \in G : x * y = y * x$$

then G is called an Abelian Group.

Notation: We denote $(*)$ by $(\cdot)$ if its multiplicative, and $(+)$ if its additive.

Proposition 1.0.1 : Let $(G, \cdot)$ be a group. then:

1. The Neutral Element is uniuqe.

2. The inverse is unique

- 3.

$$\forall x, y \in G : (x \cdot y)^{-1} = y^{-1} \cdot x^{-1}$$

- 4.

$$\forall x, y, z \in G : \begin{cases} xy = xz \\ yx = zx \end{cases} \implies \begin{cases} y = z \\ y = z \end{cases}$$

Proof. 1. Let $e_1, e_2 \in G$ be a Neutral Element, then:

$$e_1 = e_1 \cdot e_2 = e_2$$

2. let $x \in G$ and $x_1, x_2 \in G$ be its inverses, then:

$$x_1 = x_1 \cdot e = x_1 \cdot (x \cdot x_2) = (x_1 \cdot x) \cdot x_2 = e \cdot x_2 = x_2$$

3. Let $x, y' \in G$. then:

$$\begin{aligned} (x \cdot y) \cdot (x \cdot y)^{-1} &= e \implies y \cdot (x \cdot y)^{-1} = x^{-1} \\ &\implies (x \cdot y)^{-1} = y^{-1} \cdot x^{-1} \end{aligned}$$

□

Exercise

Let $(G, \cdot)$ be a group and $x_1, x_2, \dots, x_n \in G$. then:

- $(x_1 \cdots x_n)^{-1} = x_n^{-1} \cdots x_1^{-1}$
- $(x_1^{-1})^{-1} = x_1$

Definition 1.0.2 : Let $(G, \cdot)$ be a group, $n \in \mathbb{Z}$ and $x \in G$, define

$$x^n = \begin{cases} x \cdot x \cdots x & \text{if } n \geq 1 \\ e & \text{if } n = 0 \\ x^{-1} \cdot x^{-1} \cdots x^{-1} & \text{if } n \leq -1 \end{cases} \implies$$

Example:

1. $(\mathbb{Z}, +), (\mathbb{Q}^*, \cdot), (\mathbb{R}, +), (\mathbb{C}^*, \cdot)$
2. The set $\mathcal{F}(\mathbb{R}, \mathbb{R})$ with addition of maps is an Abelian Group, with the null map as Neutral Element
3. The set S_n of all bijection of $\{1, \dots, n\}$ with composition of maps is a group

Definition 1.0.3 (Sub Group) : Let $(G, \cdot)$ be a group and $H \subset G$ we say that H is a Subgroup of G if $(H, \cdot)$ is a gorup

Proposition 1.0.2 :

Let $(G, \cdot)$ a group and $H \subset G$. then H is a Subgroup of G if and only if:

1. $H \neq \emptyset$
2. $\forall x, y \in H : x \cdot y \in H$
3. $\forall x \in H : x^{-1} \in H$

Remark: The conditions (2) and (3) are equivalent to:

$$\forall x, y \in H : x^{-1} \cdot y \in H$$

Proof.

$$\forall x, y \in H : x^{-1} \cdot y \in H \implies \begin{cases} \forall x, y \in H : x \cdot y \in H \\ \forall x \in H : x^{-1} \in H \end{cases}$$

□

Notation: if H is a Subgroup of G , we denote

$$H \leq G$$

if $H \leq G$ with $H \neq G$, we call H a proper Subgroup of G and we write $H < G$

Exercise

Let $(G, \cdot)$ be a group, then the set:

$$Z(G) = \{x \in G : gx = xg, \forall g \in G\}$$

1. Prove that $Z(G) = G \iff G$ is an abelian group.

Proof. 1.

$$(\implies)$$

Suppose that G is an Abelian Group.

Let $x \in G$ and let $g \in G$, since G is an Abelian group, then $gx = xg$. then $x \in Z(G)$, then $Z(G) = G$

$$(\impliedby)$$

Suppose that $Z(G) = G$, let $x, y \in G$. then $x, y \in Z(G)$. so $\forall g \in G$:

$$\begin{cases} xg = gx \\ yg = gy \end{cases}$$

so for $g = y$, we get $xy = yx$ so G is an abelian group

2. Let $G \leq (\mathbb{Z}, +)$.

- if $G = \{0\}$. then $G = 0\mathbb{Z}$.
- if $G \neq \{0\}$, then $\exists m \in G$ with $m \neq 0$, without loss of generality. suppose $m > 0$, so $G \cap \mathbb{N} \neq \emptyset$, so $n = \min G \cap \mathbb{N}$, let $x \in n\mathbb{Z}$. then $x = kn, k \in \mathbb{Z}$, so $x \in G$. hence $n\mathbb{Z} \subset G$.

Let $x \in G$, so $\exists q, r \in \mathbb{Z}, 0 \leq r \leq n - 1$ such that $x = qn + r$. so $r = x - qn \in G$, if $r \neq 0$ then :

$$\begin{cases} r < n \\ r = G \cap \mathbb{N} \end{cases} \implies \begin{cases} r < n \\ n = \min G \cap \mathbb{N} \leq r, \end{cases} \text{ is a contradiction}$$

so $x = qn \in n\mathbb{Z}$, so $G \subset n\mathbb{Z}$

□

Proposition 1.0.3 : Let $H, K \leq G$, with G is a group. then:

$$H \cap K \leq G$$

Proof. Since $e \in H$ and $e \in K$, then $e \in H \cap K \neq \emptyset$.

Let $x, y \in H \cap K$, then :

$$\begin{cases} x, y \in H \\ x, y \in K \end{cases} \implies \begin{cases} x^{-1}, y \in H \\ x^{-1}, y \in K \end{cases} \implies \begin{cases} x^{-1} \cdot y \in H \\ x^{-1} \cdot y \in K \end{cases} \implies x^{-1} \cdot y \in H \cap K$$

□

Proposition 1.0.4 : Let $\{H_i\}_{i \in I}$ be a family of Subgroup of a group G , then :

$$\bigcap_{i \in I} H_i \leq G$$

Remark: $H \cup K$ is not always a Subgroup of G .

Proposition 1.0.5 : Let $H, K \leq G$, Then $H \cup K \leq G \iff \begin{cases} H \subset K \\ K \subset H \end{cases}$

Lecture 2

08:17 AM Thu, Oct 02 2025

Definition 1.0.4 : Let G be a group and $A \subset G$, The group spanned by A is the intersection of all SG of G containing A , i.e.:

$$\langle A \rangle = \bigcap_{H \leq G, A \subset H} H$$

Proposition 1.0.6 : $\langle A \rangle$ is the smallest SG of G containing A

Proof. $\langle A \rangle \leq G$

$$\begin{cases} e \in H, \forall H \leq G \implies \bigcap_{H \leq G} H \neq \emptyset \\ \forall x, y \in \langle A \rangle : x, y \in H, \forall H \leq G \text{ and } A \subset H \end{cases}$$

so $xy^{-1} \in H, \forall H \leq G \text{ and } A \subset H$

so $xy^{-1} \in \bigcap_{H \leq G, A \subset H} H$

Let $B \leq G$ such that $A \subset B$.

since

$$\bigcap_{H \leq G, A \subset H} H \subset B, \quad \text{so } \langle A \rangle \subset B$$

so $\langle A \rangle$ is the smallest SG of G containing A □

Proposition 1.0.7 : Let G be a group and $A \subset G$ such that $A \neq \emptyset$. Then:

$$\langle A \rangle := \left\{ x_1^{k_1} x_2^{k_2} \dots x_n^{k_n} : n \in \mathbb{N}, x_i \in A, k_i = \pm 1, i \in \{1, \dots, n\} \right\}$$

In particular if $A = \{a\}$. then by definition:

$$\langle A \rangle = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}$$

Definition 1.0.5 : Let G be a group and $A \subset G$, we say:

1. A span G if $G = \langle A \rangle$
2. G is of finite type if $G = \langle A \rangle$ and A is finite.
3. G is cyclic if $G = \langle a \rangle, a \in G$
4. G is a finite group if $|G| < \infty$, in this case we call $|G|$ the order of G .
5. The order of $x \in G$ is $|\langle x \rangle|$

Example:

1. $(\mathbb{Z}, +)$ is cyclic. Indeed, $\mathbb{Z} = \langle 1 \rangle = \{n \cdot 1 : n \in \mathbb{Z}\}$ or with -1 .
2. $(n\mathbb{Z}, +)$ is cyclic. Since $n\mathbb{Z} = \langle n \rangle = \{k \cdot n : k \in \mathbb{Z}\}$
3. $(\frac{\mathbb{Z}}{6\mathbb{Z}}, +)$ is cyclic. Since

$$\begin{aligned}\frac{\mathbb{Z}}{6\mathbb{Z}} &= \langle \bar{1} \rangle = \{n \cdot \bar{1} : n \in \{0, 1, \dots, 5\}\} \\ &= \langle \bar{5} \rangle = \{n \cdot \bar{5} : n \in \{0, 1, \dots, 5\}\}\end{aligned}$$

Proposition 1.0.8 : Any cyclic group is Abelian.

Proof. Let $G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}$, we define:

$$HK = \{hk : h \in H, k \in K\}$$

□

PRODUCT OF SG:

Let G be a group and $H, K \leq G$, we define

$$HK = \{hk : h \in H, k \in K\}$$

Remark: $H, K \subset HK$

Proposition 1.0.9 :

$$HK \leq G \implies HK = KH$$

Proof. ($\implies$) since $HK \leq G$, then $HK \neq \emptyset$, let $x \in HK$. So $x = hk$ where $h, k \in H, K$. since it's a subgroup then, $x^{-1} = k^{-1}h^{-1} \in HK$, so $HK \subset KH$, let $x \in KH$. then $x = kh$ where $k, h \in K, H$ so $x^{-1} = h^{-1}k^{-1} \in HK$. Since $HK \leq G$ then $x \in HK$ so:

$$KH \subset HK$$

□

Proposition 1.0.10 : if $HK \leq G$, then HK is the smallest SG of G containing H and K that's so:

$$HK = \langle H \cup K \rangle$$

Proof. Set

$$\begin{aligned} L &= \left\{ x_1^{k_1} \cdots x_n^{k_n} : n \in \mathbb{N}, x_i \in H \cup K, k_i = \pm 1 \right\} \\ &= \langle H \cup K \rangle \end{aligned}$$

Let $x \in HK$, then $x = hk \in L$. so $HK \subset L$, since $H, K \subset HK$. then $H \cup K \subset HK$, since $HK \leq G$, then $\langle H \cup K \rangle \subset HK$ (see definition):

$$HK = \langle H \cup K \rangle$$

□

1.1 Quotient Group

Exercise

Let $H \leq G$, G is a group and let $x, y \in G$. Show that:

1. $xH = H \iff x \in H$
2. $x^{-1}y \in H \iff y \in xH$
3. $H^{-1} = H$ with $H^{-1} = \{h^{-1} : h \in H\}$

Proof.

$$x \in H \implies H \subset xH$$

$$h \in H \implies h = xx^{-1}h = x(x^{-1}h) \in xH$$

□

- Let $H \leq G$, define on G the binary operations $\mathcal{R}_g$ and $\mathcal{R}_d$ by:

$$\forall x, y \in G : \begin{cases} x\mathcal{R}_g y \iff x^{-1}y \in H \\ x\mathcal{R}_d y \iff yx^{-1} \in H \end{cases}$$

we can show that $\mathcal{R}_g$ and $\mathcal{R}_d$ are equivalence relations, (reflexive, symmetric, transitive) let $x \in G$. that left class of x by $\mathcal{R}_g$ is:

$$\begin{aligned} \bar{x} &= \{y \in G : x\mathcal{R}_g y\} \\ &= \{y \in G : x^{-1}y \in H\} = \{y \in G : y \in xH\} = xH \end{aligned}$$

Similarly, the right class of x is:

$$\bar{x}^d = \{y \in G : x\mathcal{R}_d y\} = Hx$$

QUOTIENT OF G BY $\mathcal{R}_g$ AND $\mathcal{R}_d$:

By definition:

$$\frac{G}{\mathcal{R}_g} = \{\bar{x} : x \in G\} = \{xH : x \in G\} \stackrel{\text{def}}{=} \left(\frac{G}{H}\right)_g$$

where:

$$\frac{G}{\mathcal{R}_d} = \{\bar{x}^d : x \in G\} = \{Hx : x \in G\} \stackrel{\text{def}}{=} \left(\frac{G}{H}\right)_d$$

Proposition 1.1.1 : $\left(\frac{G}{H}\right)_g$ and $\left(\frac{G}{H}\right)_d$ are partition of G .

Proof.

$$xH \neq \emptyset \quad (x = x \cdot e \in xH)$$

$$\bigcup_{x \in G} xH = G$$

$$xH \cap yH \neq \emptyset \implies xH = yH$$

□

Proposition 1.1.2 :

1. $\left(\frac{G}{H}\right)_g$ and $\left(\frac{G}{H}\right)_d$ are equipotent.
2. $\forall x \in G : xH$ and Hx are equipotent (in bijection).

Proof. 1. Let

$$\begin{aligned} f : \left(\frac{G}{H}\right)_g &\longrightarrow \left(\frac{G}{H}\right)_d \\ \bar{x} = xH &\longmapsto f(xH) = (xH)^{-1} = Hx^{-1} \end{aligned}$$

f is well defined:

$$\begin{aligned} \bar{x} = \bar{y} &\implies f(\bar{x}) = f(\bar{y}) \\ \bar{x} = \bar{y} &\implies xH = yH \\ &\iff x^{-1}yH = H \\ &\implies Hy^{-1}x = H \\ &\implies Hy^{-1} = Hx^{-1} \implies f(\bar{x}) = f(\bar{y}) \end{aligned}$$

□

Lecture 3

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Proposition 1.1.3 :

- $\left(\frac{G}{H}\right)_d$ and $\left(\frac{G}{H}\right)_d$ are equipotent.
- and for all $x \in G : xH$ and Hx are equipotent.

Proof.

$$\begin{aligned} f : \left(\frac{G}{H}\right)_g &\longrightarrow \left(\frac{G}{H}\right)_d \\ xH &\longmapsto Hx^{-1} \end{aligned}$$

we have for all $x, y \in G$:

$$xH = yH \implies Hx^{-1} = Hy^{-1} \implies f(xH) = f(yH)$$

so f is well defined.

$$f(xH) = f(yH) \iff Hx^{-1} = Hy^{-1} \implies xH = yH \implies f \text{ injective}$$

Let $Hy \in \left(\frac{G}{H}\right)_d$ then:

$$f(y^{-1}H) = Hy \text{ with } y^{-1}H \in \left(\frac{G}{H}\right)_g$$

hence f is surjective, thus f is bijective. Let

$$\begin{aligned} f: xH &\longrightarrow Hx \\ xh &\longmapsto hx \end{aligned}$$

we have that f is bijective. □

Theorem 1.1.4 (Lagrange Theorem) : Let G be a finite group and let $H \leq G$, then $|H| \mid |G|$

Proof. Let $H \leq G$. then $\left(\frac{G}{H}\right)_g$ is a partition of G , Let $\left(\frac{G}{H}\right)_g = \{x_1H, \dots, x_nH\}$ for some $n \in \mathbb{N}$, so $G = \bigcup_{i=1}^n x_iH$. Since $\left(\frac{G}{H}\right)_g$ is a partition of G , then:

$$|G| = \left| \bigcup x_iH \right| = \sum_{i=1}^n |x_iH|$$

we have H and x_iH are equipotent for all $i \in \{1, \dots, n\}$, take

$$\begin{aligned} f: H &\longrightarrow x_iH \\ h &\longmapsto x_ih \end{aligned}$$

so

$$|G| = \sum_{i=1}^n |H| = n |H|$$

hence $|H| \mid |G|$ □

Notation:

$n = [G : H]$ called the index of H in G . if $|G| < +\infty$, then

$$[G : H] = \frac{|G|}{|H|}$$

For $G = (\mathbb{Z}, +)$ and $H = n\mathbb{Z}$ where $n \in \mathbb{N}$, we have:

$$[G : H] = \left| \left(\frac{G}{H}\right) \right| = n$$

Corollary 1.1.5 : Every finite group G of prime order is cyclic spanned any element $x \in G \setminus \{e\}$

Proof. Let $x \in G \setminus \{e\}$. By lagrange theorem we have $|\langle x \rangle| \mid |G| = p$. so $\langle x \rangle = \{e\}$ or $G = \langle x \rangle$. since $x \neq e$, then $\langle x \rangle \neq \{e\}$. So $G = \langle x \rangle$ $\square$

Remark:

The reciprical of this result is not true. we can have a cyclic group with a non prime order,

$$\frac{\mathbb{Z}}{6\mathbb{Z}} = \langle \bar{1} \rangle = \langle \bar{5} \rangle$$

is cyclic and $\left| \frac{\mathbb{Z}}{6\mathbb{Z}} \right| = 6$ not prime.

1.1.1 Normal Subgroups

Definition 1.1.1 : Let $H \leq G$. we say that H is a normal group of G if:

$$\forall x \in G : xH = Hx \quad (\text{denoted } H \triangleleft G)$$

Remark:

If G is abelian, then any subgroup of G is normal

Proposition 1.1.6 : Let $H \leq G$. then the following statments are equivalent:

$$\textcircled{1} \quad \forall x \in G : xHx^{-1} = H$$

$$\textcircled{2} \quad \forall x \in G : xHx^{-1} \subset H$$

Proof. Exercise $\square$

Let G be a group and $(H \triangleleft G)$. then:

$$\left(\frac{G}{H} \right)_g = \left(\frac{G}{H} \right)_d \stackrel{\text{def}}{=} \frac{G}{H}$$

we equip $\frac{G}{H}$ by the binary operation, defined:

$$\forall x, y \in G : (xH) \cdot (yH) = \bar{x} \cdot \bar{y} = \overline{xy} = (xy) \cdot H$$

Theorem 1.1.7 : $\left(\frac{G}{H}, \cdot \right)$ is a group called the quotient group of G by H

Proof. $\Leftrightarrow$ Associativity

$\Leftrightarrow$ The neutral element of $\left(\frac{G}{H} \right)$ is $H = \bar{e} = eH$

$$\Leftrightarrow \text{for all } \bar{x} \in \frac{G}{H} : \quad \bar{x}^{-1} = \overline{x^{-1}} = (xH)^{-1}$$

□

1.2 Group Morphism

Definition 1.2.1 : Let $(G, \cdot)$ and $(G', \mathcal{T})$ be two groups. A map $f : G \rightarrow G'$ is a group morphism if for all $x, y \in G$ we have:

$$f(x \cdot y) = f(x) \mathcal{T} f(y)$$

Proposition 1.2.1 : Let $f : G \rightarrow G'$ a Group morphism, and e, e' are the neutral elements of G and G' resp. Then:

$$\textcircled{1} \quad f(e) = e'$$

$$\textcircled{2} \quad \forall x \in G : \quad f(x^{-1}) = [f(x)]^{-1}$$

$$\textcircled{3} \quad \forall x \in G, \quad \forall n \in \mathbb{Z} : \quad f(x^n) = [f(x)]^n$$

Proof. $\boxed{1} \quad \clubsuit x = y = e \implies f(e) = f(e)f(e)$

$$\boxed{2} \quad \clubsuit y = x^{-1} \implies e' = f(x)f(x^{-1})$$

$$\boxed{3} \quad \clubsuit \text{ use induction}$$

□

Example: Let $H \triangleleft G$, then:

$$\begin{aligned} i : G &\longrightarrow G \\ h &\longmapsto h \end{aligned}$$

and

$$\begin{aligned} s : G &\longrightarrow \frac{G}{H} \\ x &\longmapsto \bar{x} = xH \end{aligned}$$

are group morphism called injective and surjective, resp.

1.2.1 Kernel-Image

Let $f : G \rightarrow G'$, The kernel of f is:

$$\text{Ker}(f) = \{x \in G : f(x) = e'\} = f^{-1}(\{e'\})$$

The image of f is:

$$\text{Im}(f) = \{f(x) : x \in G\} = f(G)$$

Proposition 1.2.2 : Let $f : G \rightarrow G'$ be a Group morphism, then:

- ① If $H \leq G$, then $f(H) \leq G'$
- ② $H' \leq G'$, then $f^{-1}(H') \leq G$

so

$$\text{Im}(f) = f(G) \leq G'$$

and

$$f^{-1}(\{e'\}) = \text{Ker}(f) \leq G$$

Proposition 1.2.3 : Let $f : G \rightarrow G'$ be a group morphism, then:

- ① f injective $\iff \text{Ker}(f) = \{e\}$
- ② f surjective $\iff \text{Im}(f) = G'$

Proof. Exercise (you know it won't happen) ☹

□

Proposition 1.2.4 : Let $H \leq G$, then:

$$H \triangleleft G \iff \exists G' \text{ a group and GM } f : G \rightarrow G' \text{ such that: } H = \text{Ker}(f)$$

Proof.

$$(\iff)$$

suppose that $\exists G'$ a group, and $f : G \rightarrow G'$ a group morphism such that $H = \text{Ker}(f)$ Let

$x \in G$, and let $y = xgx^{-1} \in x\text{Ker}(f)x^{-1}$, hence:

$$\begin{aligned} f(y) &= f(x)f(g)f(x^{-1}) \\ &= f(x)f(g)f(x^{-1}) \\ &= e' \end{aligned}$$

so $y \in \text{Ker}(f)$. look at the hand below

$$\Leftrightarrow x\text{Ker}(f)x^{-1} \subset \text{Ker}(f)$$

Therefore $H = \text{Ker}(f) \triangleleft G$.

$$(\implies)$$

Let $H \triangleleft G$, and let

$$\begin{aligned} f: G &\longrightarrow \frac{G}{H} \\ x &\longmapsto \bar{x} = xH \end{aligned}$$

we have f is a group morphism, and $\text{Ker}(f) = H$ □

Theorem 1.2.5 (First Theorem of Isomorphism) : Let $f: G \longrightarrow G'$ be a group morphism, then $\frac{G}{\text{Ker}(f)} \sim \text{Im}(f)$

Proof. $\Leftarrow$ Let

$$\begin{aligned} \tilde{f}: \frac{G}{\text{Ker}(f)} &\longrightarrow \text{Im}(f) \\ \bar{x} &\longmapsto \tilde{f}(\bar{x}) = f(x) \end{aligned}$$

Let $\bar{x}, \bar{y} \in \frac{G}{\text{Ker}(f)}$ such that $\bar{x} = \bar{y}$, then $x\text{Ker}(f) = y\text{Ker}(f)$, therefore $x^{-1}y \in \text{Ker}(f)$, hence we can deduce

$$\begin{aligned} f(x^{-1}y) = e &\iff f(x) = f(y) \\ &\iff \tilde{f}(\bar{x}) = \tilde{f}(\bar{y}) \end{aligned}$$

so f is well defined and injective. f is surjective by construction □

Lecture 4

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Decomposition of a Group Morphism:

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ \downarrow j & & \uparrow i \\ G/\text{Ker}(f) & \xrightarrow{\tilde{f}} & \text{Im}f \end{array}$$

- Put $f = i \circ \tilde{f} \circ j$
- if f is isomorphism: $f = \tilde{f}$.
- if f is injective: $f = i \circ \tilde{f}$
- if f is surjective: $f = \tilde{f} \circ j$.

where

$$\begin{array}{ccc} j: G & \longrightarrow & G/\text{Ker}f \\ x & \longmapsto & \bar{x} \\ \\ i: \text{Im}f & \longrightarrow & G' \\ y & \longmapsto & y \end{array}$$

Remark:

If $f: G \rightarrow G'$ isomorphism, then $f^{-1}: (G', *) \rightarrow (G, \cdot)$ is also isomorphism.

Theorem 1.2.6 : If $G \xrightarrow{f} G'$, then

- ① G is Abelian group $\iff G'$ is Abelian group.
- ② G is cyclic $\iff G'$ is cyclic.
- ③ $\forall x \in G : (|x| = n \iff |f(x)| = n).$
- ④ $G = \langle x \rangle \iff G' = \langle f(x) \rangle$

Proof. ♦ ① Let $a = f(x_1), b = f(x_2) \in G'$. Then

$$\begin{aligned} ab &= f(x_1)f(x_2) \\ &= f(x_1x_2) \\ &= f(x_2x_1) = f(x_2)f(x_1) = ba \end{aligned}$$

♦ ② Let $G = \langle x \rangle$. let $y \in G'$, so $\exists x_1 \in G = \langle x \rangle : y = f(x_1)$, so there exists $k \in \mathbb{Z} : y = f(x^k) = f(x)^k \in \langle f(x) \rangle$, so $G' \subset \langle f(x) \rangle$. Since $f(x) \in G'$, then $\langle f(x) \rangle \subset G'$. So $G' = \langle f(x) \rangle$.

- ❖ ③ If $|x| = n$, then $e = f(e) = f(x^n) = f(x)^n$. So $|f(x)| \mid n$. Let $m = |f(x)|$, then $e = f(x)^m = f(x^m)$, so $x^m = e$ (by injectivity). So $n \mid m$, hence $m = n$. □

Proposition 1.2.7 : If G is cyclic, then

- ① If G is finite, then $G \simeq \frac{\mathbb{Z}}{n\mathbb{Z}}$ where $|G| = n$.
- ② If G is infinite, then $G \simeq \mathbb{Z}$.

Proof. Let

$$\begin{aligned} f: \mathbb{Z} &\longrightarrow G \\ k &\longmapsto x^k, \end{aligned}$$

where $G = \langle x \rangle$. By 1st isomorphism theorem, we have $\frac{\mathbb{Z}}{\text{Ker}f} \simeq G$.

- ❶ If f is injective, then $\text{Ker}f = \{0\}$. So $\mathbb{Z}' \simeq \frac{\mathbb{Z}}{\{0\}} \simeq G$.
- ❷ If f is not injective, then $\text{Ker}f = n\mathbb{Z}$ ($\text{Ker}f \leq \mathbb{Z}$) for some $n \in \mathbb{N}$. So $\frac{\mathbb{Z}}{n\mathbb{Z}} \simeq G$. □

✎ Let $A \subset G$, G is a group. Define $C(A) = \{g \in G \mid \forall x \in A: gx = xg\} \leq G$.

✎ Let G be a group and $\mathcal{R}$ be the binary relation defined by:

$$\forall x, y \in G: x\mathcal{R}y \iff \exists g \in G: y = gxg^{-1}$$

we have $\mathcal{R}$ is an equivalence relation and if $x\mathcal{R}y$ then x and y are conjugates.

✎ Define

$$\begin{aligned} \Omega(x) &= \{y \in G \mid x\mathcal{R}y\} \\ &= \left\{ y \in G \mid y = gx^{-1}g, \quad g \in G \right\} \\ &= \left\{ gxg^{-1} \mid g \in G \right\} = \Omega_x. \end{aligned}$$

Ω_x is called the conjugacy class of x , also the orbit of x .

- ❖ Let $\frac{G}{\mathcal{R}} = \{\Omega_x \mid x \in G\}$. We have $\frac{G}{\mathcal{R}}$ is a partition of G . Suppose $\Omega_x \cap \Omega_y \neq \emptyset$. So $\exists z \in \Omega_x$ and $z \in \Omega_y$. so

$$\begin{cases} z = gxg^{-1} \\ z = g_1yg_1^{-1} \end{cases} \quad g, g_1 \in G.$$

So $x = g^{-1}g_1yg_1^{-1}g = (g^{-1}g_1)y(g^{-1}g_1)^{-1} \in \Omega_y$. Then $\Omega_x \subset \Omega_y$.

Theorem 1.2.8 : For all $x \in G$, $|\Omega_m| = [G : C(x)] = \left| \left(\frac{G}{C(x)} \right)_g \right|$

Proof.

$$\begin{aligned} f : \Omega_x &\longrightarrow \frac{G}{C(x)} \\ gx^{-1}g &\longmapsto \bar{g} = gC(x) \end{aligned}$$

f is well defined.

$$\begin{aligned} gxg^{-1} = g_1xg_1^{-1} &\implies (g_1^{-1}g)x(g_1^{-1}g)^{-1} = x &\implies g_1^{-1}g \in C(x) \\ &\implies g \in g_1C(x) \\ &\implies gC(x) \subset g_1C(x) \end{aligned}$$

f injective:

$$\begin{aligned} f(gxg^{-1}) = f(g_1xg_1^{-1}) &\implies gC(x) = g_1C(x) \\ &\implies g^{-1}g_1C(x) = C(x) \\ &\implies g^{-1}g_1 \in C(x) \\ &\implies (g^{-1}g_1)x = x(g^{-1}g_1) \\ &\implies g_1xg_1^{-1} = gxg^{-1} \end{aligned}$$

Remark:

For all $x \in G$: $(\Omega_x = \{x\}) \iff x \in Z(G)$.

$$\begin{aligned} x \in Z(G) &\iff C(x) = G \\ &\iff \left| \left(\frac{G}{C(x)} \right)_g \right| = \left| \frac{G}{G} \right| = 1 \\ &\iff |\Omega_x| = 1 \\ &\iff \Omega_x = \{x\}. \end{aligned}$$

□

Proposition 1.2.9 : Let G be a finite group of order $n \geq 2$. Then

$$|G| \text{ is prime} \iff \{e\} \text{ and } G \text{ are the only SG of } G.$$

Proof. $\Leftrightarrow (\implies)$ use LT.

$\Leftrightarrow (\impliedby)$ Let $|G| = n = mp$, p prime. Let $x \in G \setminus \{e\}$, so $\langle x \rangle = G$. Let $y = x^m \neq e$ since $p \geq 2$, so $\langle y \rangle = G$. Then $|G| = |\langle y \rangle| = p = n$. $\square$

Theorem 1.2.10 : Every group of order p^2 is abelian group, p prime.

Proof. Let $|G| = p^2$. By LT, $|Z(G)| \mid p^2$. So $|Z(G)| \in \{1, p, p^2\}$.

❶ If $|Z(G)| = 1$, then $Z(G) = \{e\}$.

❷ If $|Z(G)| = p$, then

$$\left| \frac{G}{Z(G)} \right| = \frac{|G|}{|Z(G)|} = p.$$

❸ So $|Z(G)| = p^2$. Then $G = Z(G)$. $\square$

Theorem 1.2.11 (Cauchy's Theorem) : Let G be a group of order n . Let p be prime such that $p \mid n$. Then $\exists x \in G$ where $|x| = p$.

Proof. Let $|G| = np$, where p prime. (to be continued). $\square$

Any group of order $p, p^2, \dots$, $Z(G)$ cant be $\{e\}$.

Chapter 2

Rings

Lecture 5

08:06 AM Thu, 0ct 23 2025

$(A, +, \cdot)$ is a ring if:

① $(A, +)$ is an abelian group.

② Is associative: $\forall x, y, z \in A : (xy)z = x(yz)$.

③ Is distributive: $\forall x, y, z \in A :$

$$\begin{cases} x(y + z) = xy + xz, \\ (x + y)z = xz + yz. \end{cases}$$

④ The unity element: $\exists e \in A : x \cdot e = e \cdot x = x$.

⑤ Is commutative: $\forall x, y \in A : xy = yx$.

Example:

① $\mathbb{Q}, \mathbb{Z}, \mathbb{R}, \mathbb{C}$ are rings.

③

$$\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}, i^2 = -1\}$$

③ $\mathbb{R}[X]$.

Definition 2.0.1 : A subring B of a ring A is a subgroup of $(A, +)$ if

$$1 \in B, \forall x, y \in B : xy \in B.$$

Definition 2.0.2 (Rings Product) : Let A, B be two rings. We define the multiplication in $A \times B$ by:

$$\forall (a, b), (c, d) \in A \times B : (a, b)(c, d) = (ac, bd).$$

Then $(A \times B, +, \cdot)$ is a ring with unity $(1_A, 1_B)$.

Let A be a ring and $a, b \in A$.

① We say that b divides a (a is a multiple of b) if

$$\exists q \in A : a = qb.$$

② a is invertible if $\exists a' \in A : aa' = 1$.

The set of invertible elements of A is denoted by A^* .

Remark:

If B is a subgroup of A , then an element A of B can be invertible in A but not in B ($2 \in \mathbb{Z} \subset \mathbb{Q}$).
for all $x \in A$:

$$x \cdot 0 = x(0 + 0) = x \cdot 0 + x \cdot 0,$$

thus,

$$0 = x \cdot 0.$$

Proposition 2.0.1 : $(A^*, \cdot)$ is an abelian group with 1_A as neutral element.

Proposition 2.0.2 :

$$(A \times B)^* = A^* \times B^*$$

Proof. We have:

$$\begin{aligned} (a, b) \in (A \times B)^* &\iff \exists (x, y) \in A \times B : (a, b)(x, y) = (ax, by) = (1_A, 1_B). \\ &\iff a \in A^* \wedge b \in B^* \\ &\iff (a, b) \in A^* \times B^*. \end{aligned}$$

□

Definition 2.0.3 : A field K is a ring with $K^* = K \setminus \{0\}$.

Example:

- ① $\mathbb{Z}^* = \{\pm 1\} \implies \mathbb{Z}$ is not a field.
- ② $(\mathbb{R}[X])^* = \mathbb{R}^*$.
- ③ $(\mathbb{Z}[i])^* = \{\pm 1, \pm i\}$.

2.1 Ring Morphism

Definition 2.1.1 : $f : A \longrightarrow B$ a ring morphism if:

$$\forall x, y \in A : \begin{cases} f(x + y) = f(x) + f(y), \\ f(x \cdot y) = f(x)f(y), \\ f(1_B) = 1_B, \end{cases}$$

Definition 2.1.2 (Ideals) : An ideal $I \subset A$ is a subgroup $(I, +)$ of $(A, +)$, if

$$\forall x \in A, \forall a \in I : xa \in I.$$

✧ NOTATION: $(a) = aA = \{ab : b \in A\}$ is an ideal of A spanned by a .

Definition 2.1.3 : I is a principal ideal of A if $\exists a \in A$ such that $I = (a) = aA$.

Example: If $f : A \rightarrow B$ a ring morphism, then

$\text{Ker}(f) = \{x \in A : f(x) = 0_B\}$ is an ideal of A and $\text{Im}(f)$ is subring of B .

Proposition 2.1.1 : A ring A is a field if and only if $\{0\}$ and A are the only ideals of A .

Proof.

($\Rightarrow$)

Let $I \neq \{0\}$ be an ideal of A . Then there exists $a \in I \setminus \{0\} \subset A^*$. So $1 = a^{-1}a \in I$. Then for all $x \in A : x = x \cdot 1 \in I$, so $I = A$.

($\Leftarrow$)

Let $a \in A \setminus \{0\}$. Then $(a) = A$, since $0 \neq a \in (a)$. Then $1 = ab, b \in A$. So $a \in A^*$. So $A^* = A \setminus \{0\}$ so A is a field. $\square$

Remark:

$f : K \rightarrow B$ a ring morphism with K is a field. Since $\text{Ker}(f) = \{0\}$ ($\text{Ker}(f) \neq \{0\}$), then f is injective, $\text{Im}(f)$ is a field.

2.2 Quotient Ring

Let A be a ring and I not included in A an ideal, since $(I, +)$ proper subgroup $(A, +)$, then $\frac{A}{I}$ is an abelian group, we define $\frac{A}{I}$ the binary operation, by:

$$\forall \bar{x}, \bar{y} \in \frac{A}{I} : \bar{x} \cdot \bar{y} = \overline{xy}.$$

Then binary operation is well defined. Indeed:

$$\begin{cases} \bar{x} = \bar{x}' \\ \bar{y} = \bar{y}' \end{cases} \implies \begin{cases} x - x' \in I \\ y - y' \in I \end{cases} \implies \begin{cases} x' = x + a, a \in I \\ y' = y + b, b \in I. \end{cases}$$

which implies that

$$\overline{x'y'} \implies \overline{x'y'} = \overline{x+a} \cdot \overline{y+b} = \overline{x \cdot y} = \overline{xy}.$$

Proposition 2.2.1 : $(\frac{A}{I}, +, \cdot)$ is a ring called the quotient ring of A by I .

Proof.

$$(\overline{xy})\bar{z} = (\overline{xy})\bar{z} = \overline{(xy)z} = \overline{x(yz)} = \bar{x}(\overline{yz}) = \bar{x}(\bar{y}\bar{z})$$

□

Theorem 2.2.2 (First theorem of isomorphism (ring version)) : Let $f : A \rightarrow B$ be a ring morphism. Then

$$\frac{A}{\text{Ker}(f)} \simeq \text{Im}(f)$$

Proof. Let

$$\begin{aligned} \tilde{f} : \frac{A}{\text{Ker}(f)} &\longrightarrow \text{Im}(f) \\ \bar{x} &\longmapsto \tilde{f}(\bar{x}) = f(x). \end{aligned}$$

$$\bar{x} = \bar{y} = x - y \in \text{Ker}(f)$$

$$\iff f(x - y) = 0$$

$$\iff f(x) = f(y)$$

$$\iff \tilde{f}(\bar{x}) = \tilde{f}(\bar{y}) \iff \tilde{f} \text{ is well defined and injective.}$$

□

Lecture 6

08:05 AM Thu, Nov 06 2025

Theorem 2.2.3 (Chinese remainder theorem) : Let $p, q \in \mathbb{N}, (p, q) = 1$. Then $\mathbb{Z}_{pq}$ is isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_q$.

Proof. Let

$$\begin{aligned} f : \mathbb{Z} &\longrightarrow \mathbb{Z}_p \times \mathbb{Z}_q \\ k &\longmapsto (\bar{k}^p, \bar{k}^q) \end{aligned}$$

We have f is a RM, also,

$$\begin{aligned} \text{Ker}(f) &= \{k \in \mathbb{Z} : \bar{k}^p = \bar{0}^p \text{ and } \bar{k}^q = \bar{0}^q\} \\ &= \{k \in \mathbb{Z} : k \in p\mathbb{Z} \cap q\mathbb{Z}\} \\ &= \text{lcm}(p, q)\mathbb{Z} = pq\mathbb{Z}. \end{aligned}$$

Hence the kernel of f is $\{\bar{0}^{pq}\}$. Then by first isomorphism theorem,

$$\mathbb{Z}_{pq} \text{ is isomorphic to } \text{Im}(f) \subset \mathbb{Z}_p \times \mathbb{Z}_q.$$

so $pq = |pq\mathbb{Z}| = |\text{Im}(f)| = |\mathbb{Z}_p \times \mathbb{Z}_q|$. Then $\mathbb{Z}_{pq}$ is isomorphic to $\text{Im}(f) \times \mathbb{Z}_p \times \mathbb{Z}_q$. $\square$

Proposition 2.2.4 : Let C_p and C_q be two cyclic groups. Then $C_p \times C_q$ is cyclic if and only if $(p, q) = 1$.

Proof. $\blacklozenge (\Leftarrow)$: $(p, q) = 1 \iff C_p \times C_q$ is isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_q$ which is isomorphic to $\mathbb{Z}_{pq}$. Thus, its cyclic.

$\blacklozenge (\Rightarrow)$: $d = (p, q) > 1$. Then $\text{lcm}(p, q) = a < pq$. Let $(\bar{x}^p, \bar{x}^q) \in C_p \times C_q$ which is isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_q$. Then $(\bar{x}^p, \bar{y}^q) = (\bar{a}\bar{x}^p, \bar{a}\bar{y}^q) = (\bar{0}^p, \bar{0}^q)$. So $C_p \times C_q$ is not cyclic. $\square$

2.2.1 Euler function

Euler function: for $n \in \mathbb{N}$, define the function φ by:

$$\begin{aligned} \varphi : \mathbb{N} &\longrightarrow \mathbb{N} \\ n &\longmapsto \varphi(n) = |\{1 \leq k \leq n : (k, n) = 1\}| \end{aligned}$$

*** Properties.** If p is prime, $\varphi(p) = p - 1$. for all $r \in \mathbb{N}$, we have,

$$\varphi(p^r) = p^r - p^{r-1} = p^{r-1}(p - 1).$$

If $n, m \in \mathbb{N}$ such that $(n, m) = 1$, then

$$\varphi(nm) = \varphi(n)\varphi(m).$$

Proof.

$$\begin{aligned} \varphi(p^r) &= |\{1 \leq k \leq p^r : (k, p^r) = 1 \iff (k, p) = 1\}| \\ &= p^r - |\{1 \leq k \leq p^r : (k, p) = p\}| \\ &= p^r - p^{r-1} \end{aligned}$$

We can use the chinese remainder theorem,

$$\begin{aligned} (n, m) &\implies 1 \implies \mathbb{Z}_{nm} \text{ is isomorphic } \mathbb{Z}_n \times \mathbb{Z}_m \\ &\implies (\mathbb{Z}_{nm})^* \text{ is isomorphic } (\mathbb{Z}_n \times \mathbb{Z}_m)^* \\ &= \mathbb{Z}_n^* \times \mathbb{Z}_m^* \end{aligned}$$

Alternative way, we have $\varphi(nm) = |(\mathbb{Z}_{nm})^*| = |\mathbb{Z}_n^* \times \mathbb{Z}_m^*| = |\mathbb{Z}_n^*| \cdot |\mathbb{Z}_m^*| = \varphi(n) \times \varphi(m)$. $\square$

Let $n \geq 2$ (where $n \in \mathbb{N}$).

✿ **Proposition.**

- ① Let $a \in \mathbb{Z}$, if $(a, n) = 1$ then $a^{\varphi(n)} = 1 [n]$.
- ② Every cyclic group of order n has $\varphi(n)$ generators.

Proof. ♦ ①. Let $a \in \mathbb{Z}$, and suppose that $(a, n) = 1$, that is;

$$\begin{aligned}
 (a, n) = 1 &\implies \exists \alpha, \beta \in \mathbb{Z} : a\alpha + \beta n = 1, \\
 &\implies \bar{a}\bar{a} = \bar{1} \in \mathbb{Z}_n, \\
 &\implies \bar{a} \in (\mathbb{Z}_n)^*, \\
 &\implies \bar{a}^{\varphi(n)} = \bar{1} \quad (|\mathbb{Z}_n^*| = \varphi(n)), \\
 &\implies a^{\varphi(n)} = 1 [n].
 \end{aligned}$$

♦ ②. Left as an exercise *. This finishes the proof, as required. $\square$

We denote the finite field $\mathbb{Z}_p := \mathbb{F}_p$.

Proposition 2.2.5 : Let $n \geq 2$ (where $n \in \mathbb{N}$). Then $\mathbb{Z}_n$ is a field if and only if n is a prime.

Proof. If n is prime, then $|\mathbb{Z}_n|^* = \varphi(n) = n - 1$. So $\mathbb{Z}_n$ is a field. Suppose $\mathbb{Z}_n$ is a field. Let $1 \leq d < n$ such that $d|n$. So there exists $q \in \mathbb{N}$ such that $n = qd$. Then $\bar{q}\bar{d} = \bar{0} \in \mathbb{Z}_n$; Since $d \neq 0$, then $\bar{d} \in (\mathbb{Z}_n)^*$. So $\bar{q} = \bar{0}$, thus $q = n$ and $d = 1$. Hence n is prime. $\square$

Theorem 2.2.6 (Fermat's Theorem) : Let p be prime. Let also $a \in \mathbb{Z}$, then

$$a^p = a [p].$$

Proof. Since $\mathbb{Z}_p$ is a field, then for all $a \in \mathbb{Z} \setminus p\mathbb{Z}$, we have: $\square$

$$\bar{a}^{p-1} = \bar{1} \in (\mathbb{Z}_p)^* \implies \bar{a}^p = \bar{a}.$$

If $a \in p\mathbb{Z}$, then $\bar{a}^p = \bar{0}^p = \bar{0} = \bar{a}$. Thus

$$a^p = a[p], \quad (\forall a \in \mathbb{Z}).$$

Definition 2.2.1 : Let R be a ring. We say that R is an integral domain (ID) if:

- ① R is commutative.
- ② R is unity.
- ③ R has no zero divisors; That is, for $a, b \in R$:

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

2.3 Principal ring

Definition 2.3.1 : A ring R is a principal ring if R is an integral domain, and every ideal of R is principal. That is; for all $I \subset R$, if I is an ideal, then:

$$I = (a) = aR, \quad (a \in R).$$

Example:

- ① $\mathbb{Z}$ is a principal ring since $n\mathbb{Z} = (n)$ (where $n \in \mathbb{N}$), which are the ideals of $\mathbb{Z}$.
- ② If $\mathbb{K}$ is a field, then $\mathbb{K}[X]$ is a principal ring. Hint: (Use the Euclidean division for the proof)..
- ③ The ring $\mathbb{Z}[X]$ is not a principal ideal since $(2, X)$ is not a principal ideal of $\mathbb{Z}[X]$.
 $((a, b) = aR + bR).$

Definition 2.3.2 : Let $a, b \in R$. Suppose that $a = bc$ for some $c \in R^*$, then we say that a and b are associates.

Remark. The relation (a and b are associates where $a, b \in R$) is an equivalence relation on R .

Proposition 2.3.1 : Let R be a ring. Suppose that R is an integral domain, then

$$aR \subset bR \iff b \mid a, \quad (a, b \in R),$$

and

$$aR = bR \iff a = bc, \quad (c \in R).$$

2.3.1 Irreducible and prime elements

Definition 2.3.3 : Let R be a ring. Suppose that R is an integral domain, an element $a \in R \setminus \{0\}$ is said to be irreducible if $a \notin R^*$ (not invertible), and

$$a = uv \implies u \in R^* \text{ or } v \in R^*, \quad (u, v \in R).$$

Example:

- The primes of $\mathbb{Z}$ are irreducible. Let p be a prime number, and let also $a, b \in \mathbb{Z}$ such that

$$\begin{aligned} p = ab &\implies a = p \text{ or } b = p \\ &\implies b = 1 \in \mathbb{Z}^* \text{ or } a = 1 \in \mathbb{Z}^* \end{aligned}$$

- Let $\mathbb{K}$ be a field. A polynomial $P \in \mathbb{K}[X]$ of degree n (where $n \in \mathbb{N}_0$) is irreducible if and only if $n \geq 1$ and the only divisions are those of degree 0 or n .
- In $\mathbb{Z}[i]$, we have $(3 + 2i)(3 - 2i) = 13$. ($\mathbb{Z}[i]^* = \{\pm 1, \pm i\}$). So 13 is not irreducible in $\mathbb{Z}[i]$, even though 13 is irreducible in $\mathbb{Z}$.

Definition 2.3.4 : Let R be a ring. Suppose that R is an integral domain and let $p \in R$ such that $p \notin R^* \cup \{0\}$. If

$$p \mid ab \implies p \mid a \text{ or } p \mid b, \quad (a, b \in R).$$

Then we say that p is prime.